

AI, Effort, and Fairness: An Ultimatum Game Experiment

Kabir Bhalla, Kanav Nanda, Siddhant Kartikeyan

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When money is earned with AI assistance rather than human effort, do people treat it as equally deserved?

Research Question & Motivation

Core Research Question

Does using AI to earn money feel more like genuine work, or more like a lucky windfall?

Specific Inquiries

- How does replacing solitary human effort with an AI assistant during the money-earning phase alter a proposer's final offer amount?
- Do responders demand a higher share when they know the proposer used AI, viewing the funds as “easy money”?
- Does algorithmic social proof have a stronger anchoring effect on AI-assisted proposers than on manual-effort proposers?

Motivation

Generative AI is reshaping the modern workplace. Because AI sits ambiguously between hard work and pure luck, it remains unclear whether AI-assisted income is perceived as fully deserved.

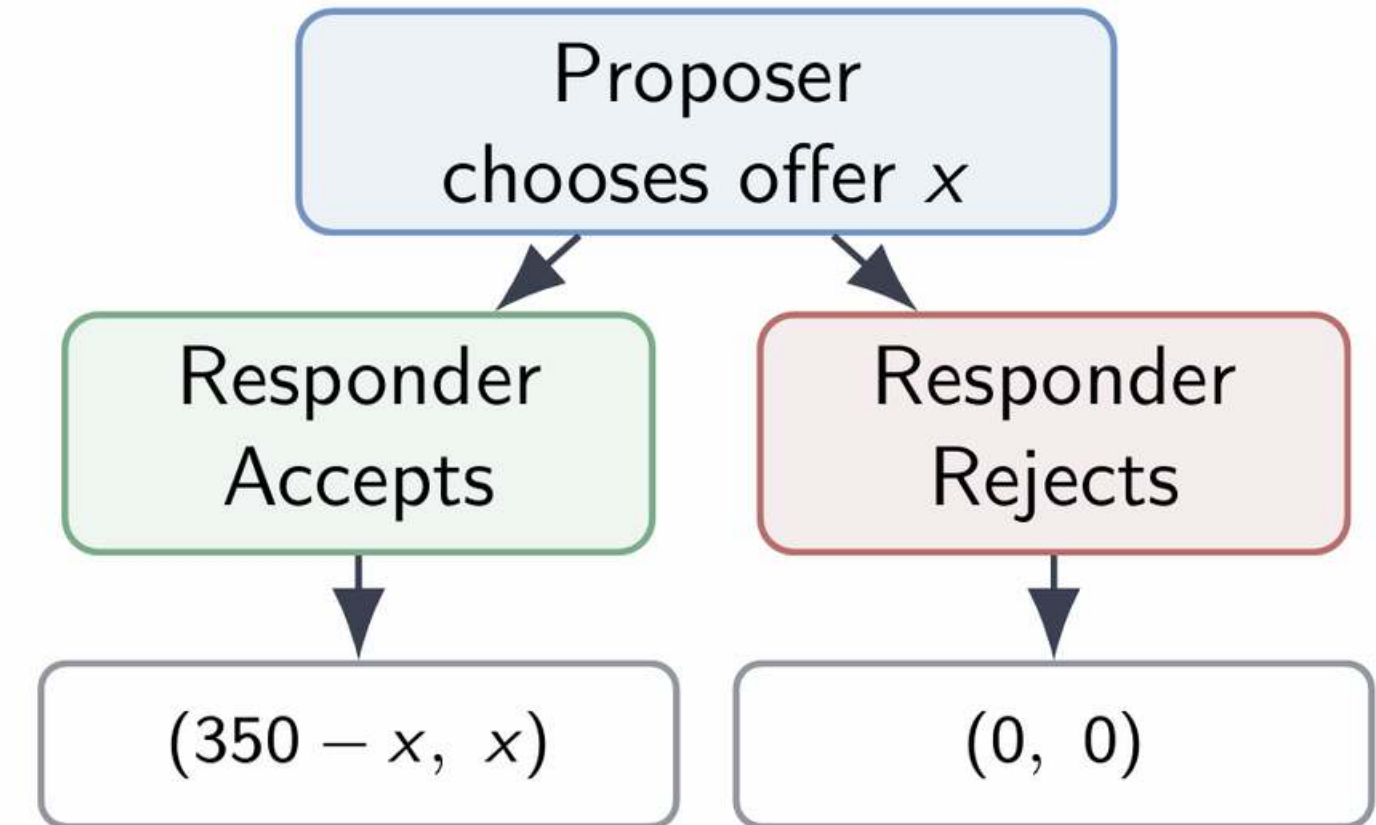
Theoretical Context: The Ultimatum Game

Roles and Actions

Role	Action
Proposer	Completes an effort task to unlock Rs. 350, then offers amount x .
Responder	Accepts or rejects the proposed split via strategy method.

Payoff Rule

Decision	Proposer	Responder
Accept	$350 - x$	x
Reject	0	0



What the Game Captures

Unlike standard backward induction, which predicts proposers offer the minimum, this setup captures fairness norms, entitlement, negative reciprocity, and costly rejection of unfairness.

3. Literature Review I: Ultimatum Game & Entitlement

Ultimatum Game Baseline

- The original Ultimatum Game shows that responders reject low offers even when rejection is costly (Güth, Schmittberger & Schwarze, 1982).
- This makes the game a clean test of fairness norms, negative reciprocity, and costly punishment.

Earned Money Changes Allocation

- Property-rights framing changes bargaining behaviour when proposers are treated as having earned stronger claims over the pie (Hoffman et al., 1994).
- Earned endowments lead allocators to keep more than randomly assigned endowments (Cherry, Frykblom & Shogren, 2002).

Takeaway for our design: If income feels personally earned, proposers should feel more entitled to keep it. Our experiment asks whether AI assistance weakens that entitlement while producing the same pie.

Design implication: the amount stays constant; the perceived source of the money changes.

4. Literature Review II: Social Information & the AI Gap

What Social-Influence Work Adds

- People use others' behaviour as informational and normative guidance, especially when the correct choice is ambiguous (Cialdini & Goldstein, 2004).
- Peer-popularity information can shift decisions and create path-dependent outcomes even when quality is fixed (Salganik, Dodds & Watts, 2006).

What It Does *Not* Test

- Earned-income studies compare effort versus windfall, but not AI-assisted earning.
- Social-proof studies test peer influence, but not whether it is stronger when entitlement is weakened.
- Ultimatum studies rarely tell responders exactly *how* the proposer earned the stake.

Resulting Gap: No design in our review jointly crosses **effort source** (human vs. AI-assisted) with **algorithmic social proof**, while also informing responders how the money was earned. AI is therefore treated as a middle category: not pure windfall, but not fully manual effort either.

Including AI in the Picture

Why AI Creates a New Fairness Problem

AI is neither pure hard work nor pure luck. It helps produce value quickly, but it may weaken the perceived personal effort behind that value.

Our Extension

We compare purely human effort with AI-assisted effort inside the same Ultimatum Game framework, then test whether algorithmic social proof pushes offers closer to a 50–50 split.

Central Mechanism

AI assistance may lower perceived entitlement, which can increase generosity and make social proof more persuasive.

Novelty Relative to the Literature

Feature	Prior Literature	Our Experiment
Source of money	Often fixed, gifted, or earned through human effort.	Proposer earns the pie under human or AI-assisted labour.
Effort source	Human effort and windfalls studied separately.	AI is positioned between hard work and easy money.
Social information	Social norms may be studied separately.	Algorithmic social proof is crossed with effort source.
Responder information	Offer amount is usually central.	Responder learns how the money was earned.
Core mechanism	Entitlement, fairness, rejection.	AI-mediated entitlement and social-proof anchoring.

Key Novelty Claim

We isolate how AI-mediated “easy money” affects entitlement and bargaining behaviour, and test whether this interacts with algorithmic social proof.

Research Hypotheses & Expected Outcomes

H1: AI-Assisted Earning Increases Offers

Proposers who use a generative AI tool to earn their money will offer a higher and more equitable amount to responders compared to those using solitary human effort. *Why?* Because they exerted less effort, their perceived entitlement decreases, treating it more like a windfall.

H2: Responders Are Stricter Toward AI

Responders will demand a higher minimum acceptable offer and reject unfair offers more often if informed that the proposer used AI assistance. *Why?* Because they view the proposer's income as "easy money," lowering their tolerance for inequality.

H3: AI Amplifies Social Proof

Proposers in the AI-assisted condition will be more easily influenced by algorithmic social proof. *Why?* Advice highlighting equal peer behaviour will anchor offers closer to a 50–50 split because the initial sense of entitlement is already weakened by the AI assistance.

Expected Direction

Manual effort strengthens entitlement; AI assistance weakens entitlement; social proof becomes highly persuasive when entitlement is ambiguous.

Experimental Design

Design Structure

2×2 between-subjects factorial design. Each proposer-responder pair is randomly assigned to one independent treatment cell to prevent cross-treatment anchoring.

Independent Variables

Factor	Levels
Source of Effort	Human Effort vs. AI-Assisted Effort
Social Proof	Present vs. Absent / Control

Treatment Matrix

	Social Proof Present	Social Proof Absent
Human	Human / Present	Human / Absent
AI	AI / Present	AI / Absent

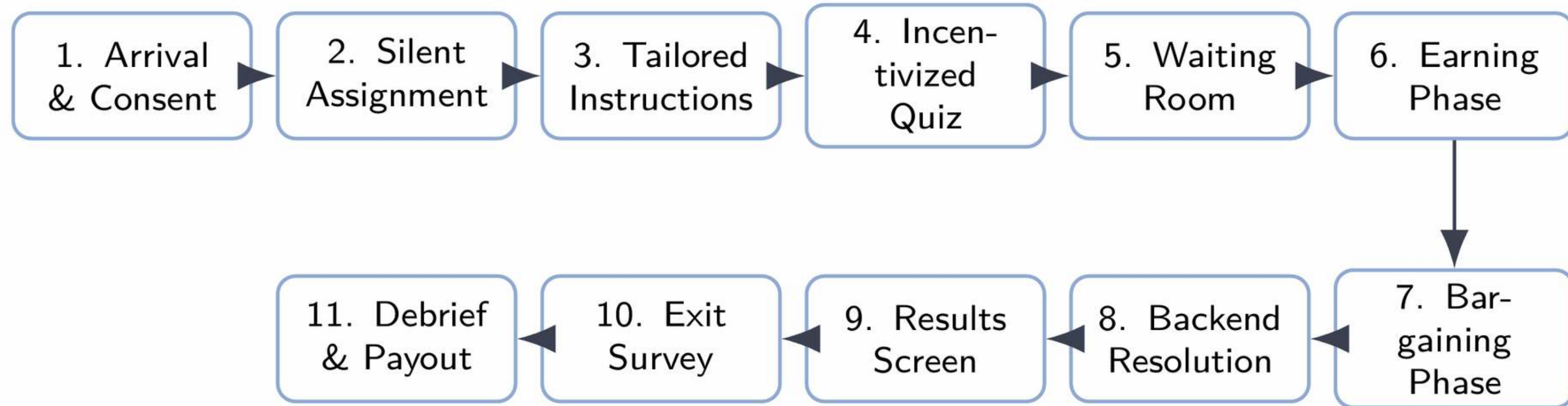
Core Environment

- **One-Shot Game:** Isolates fairness from reputation building.
- **Pie:** Rs. 350
- **Base Fee:** Rs. 100 show-up fee.
- **Incentive:** Rs. 10 bonus per correct answer on a pre-game comprehension quiz.

Strategy Method

Responders state their accept/reject preference for every Rs. 10 increment from Rs. 0 to Rs. 350 before seeing the offer, yielding exact fairness thresholds.

Treatment Manipulation & Experimental Flow



The Earning Phase Manipulation

- **Human:** Count the number of 7s in a dense matrix, compute $3 \times \sqrt{(n + 273)} \times 4 - 371$, and submit to unlock Rs. 350.
- **AI:** Click a button; the exact same task is completed instantly to unlock Rs. 350.

Resolution Logic

The software matches the proposer's actual offer to the responder's pre-submitted strategy matrix for that exact amount, computes the game payoff, then adds the quiz bonus and show-up fee.

O TREE DEMONSTRATION

Proposer Screens

- **Human Condition:** Show the 5-step counting and math matrix they must solve manually.
- **AI Condition:** Show the instant “Use AI Assistant” button interface.
- **Social Proof:** Point out where the algorithmic nudge, “peers offer Rs. 175,” appears.

Responder Screens

- **Information Phase:** Show the text detailing if the proposer used AI or human effort.
- **Strategy Matrix:** Show the UI for accepting or rejecting each Rs. 10 increment from Rs. 0 to Rs. 350.

Addressing Feedback & Study Limitations

Addressing Peer Review Feedback

Literature & Design Adjustments

- **Tightened RQs:** Re-worded research questions to directly align with the 2×2 experimental design.
- **Deepened Literature Context:** Expanded literature discussion to better engage with entitlement effects and algorithmic advice.
- **Improved Effort Task:** Upgraded the baseline effort task from merely “counting 7s” to a complex multi-step math equation.
- **Clarified Responder Action:** Added explicit detail on the strategy method mechanics.

Design Constraints

Limitations

- ① **Skill Variance:** The math task requires calculating square roots, which may create skill bias rather than pure effort.
- ② **Hidden Time Variable:** Responders do not know how long the manual worker took.
- ③ **Simulated Task:** A simulated AI math shortcut may not fully capture real-world generative AI integration.

Future Scope & Implementation

Technical Expansions

Live API Integration

Future iterations of this experiment will replace the simulated AI button with a live API integration, such as Google Gemini. Instead of an instant click, participants will have to exert “prompt engineering effort,” allowing us to study how interacting with real LLMs affects entitlement compared to basic automation.

Dynamic Social Proof

Rather than a static Rs. 175 nudge, the oTree software can be programmed to calculate the real-time running average of previous participants' offers and display that as the algorithmic advice, creating a more organic social proof environment.

Experimental Execution

Live Lab Deployment

- Deploy the finalized oTree code on a local server.
- Conduct the physical lab sessions at Plaksha University using networked terminals.
- Process all payouts using UPI QR codes at the debrief station.

Broader Demographics

While this pilot uses university students, future studies should deploy this online, for example via Prolific, to capture a broader labor force. Observing how older workers with established labor routines react to AI-assisted income will increase external validity.

Measures & Empirical Specification

Outcome Variables

- Offer amount: Rs. 0 to Rs. 350
- Share of pie offered
- Acceptance / rejection thresholds from strategy matrix
- Fairness rating: 1–7 Likert scale
- Pre-game quiz score: 0–5

OLS Specification: Proposers

$$\begin{aligned} Offer_i = & \beta_0 + \beta_1 AI_i + \beta_2 Proof_i \\ & + \beta_3 (AI_i \times Proof_i) + \varepsilon_i \end{aligned}$$

- AI_i : Dummy for AI condition.
- $Proof_i$: Dummy for social proof condition.
- β_3 : Interaction effect, testing H3.

Data Cleaning Control

Because the comprehension quiz allows participants to proceed regardless of score, participants who score 0 or 1 out of 5 will be dropped from the final analysis to ensure data validity.